Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-1 : First Order ODEs

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1. Sketch the direction field and draw the isoclines corresponding to the slopes $m = -2, -1, 0, 1, 2$.
   (i) $y' = y(2 - x)$ (ii) $y' = x^2 - y$ (iii) $y' = (x + 1)(y - 5)$.

2. Test the following equations for exactness and solve it if it is exact:
   (i) $(x + \frac{2}{y})dy + ydx = 0$ (ii) $(y - x^3)dx + (x + y^3)dy = 0$ (iii) $e^y dx + (xe^y + 2y)dy = 0$
   (iv) $(\sin x \tan y + 1)dx + \cos x \sec^2 y dy = 0$ (v) $dx = \dfrac{y}{1 - x^2y^2}dx + \dfrac{x}{1 - x^2y^2}dy$
   (vi) $-\frac{1}{y}\sin\frac{x}{y}dx + \frac{x}{y^2}\sin\frac{x}{y}dy = 0$ (vii) $(2xy^4 + y\cos x)dx + (3x^2y^2 + \sin x)dy = 0$
   (viii) $\dfrac{ydx + xdy}{1 - x^2y^2} + xdx = 0$ (ix) $\cos x \cos^2 y dx + 2\sin x \sin y \cos y dy = 0$
   (x) $(y^2e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$

3. Under what circumstances will the equation $M(x, y)dx + N(x, y)dy = 0$ have an integrating factor that is a function of the sum (i) $z = x - y$ (ii) $z = x^2 - 3y$?

4. Solve the following equations by finding an integrating factor:
   (i) $(3x^2 - y^2)dy - 2xydx = 0$ (ii)$x^2ydx - (x^3 + y^3)dy = 0$ (iii) $(x + 2)\sin ydx + x\cos ydy = 0$
   (iv) $(y\log y - 2xy)dx + (x + y)dy = 0$ (v) $(y^2 + xy + 1)dx + (x^2 + xy + 1)dy = 0$
   (vi)$(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$ (vii)$(x^3 + xy^4)dx + 2y^3dy = 0$ (viii)$(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$ (ix) $(x^2y^2 + xy + 1)ydx - (x^2y^2 - xy + 1)xdy = 0$

5. Solve the following by converting them into the form $dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy$:
   (i) $xdy - ydx = (1 + y^2)dy$ (ii) $ydx - xdy = xy^3dy$ (iii) $xdy = (x^5 + x^3y^2 + y)dx$
   (iv) $(x + y)dy = (y - x)dx$ (v) $(y^2 - y)dx + xdy = 0$ (vi) $xdy + ydx = \sqrt{xy}dy$
   (vii) $dy + \frac{y}{x}dx = \sin xdx$ (viii) $x\dfrac{dy}{dx} - y = x\sqrt{x^2 + y^2}$ (ix) $(xy^2 - e^{1/x^3})dx - x^2ydy = 0$
   (x) $xdx + ydy = xdy - ydx$

6. Show that the equation $(3y^2 - x)dx + 2y(y^2 - 3x)dy = 0$ admits an integrating factor which is a function of $(x + y^2)$. Hence solve the differential equation.

7. Solve the following linear equations:
   (i) $xy' - 3y = x^4$ (ii) $y - x + xy\cot x + xy' = 0$ (iii) $y' + y = 2xe^{-x} + x^2$
   (iv) $(x\log x)y' + y = 3x^3$ (v) $y' + y\cot x = 2x\ cosecx$ (vi) $(y - 2xy - x^2)dx + x^2dy = 0$
   (vii) $x\cos x\dfrac{dy}{dx} + y(x\sin x + \cos x) = 1$ (viii) $\dfrac{dy}{dx} + \frac{n}{x}y = \frac{a}{x^n}$

8. Apply suitable transformation to convert the following non-linear differential equations into linear differential equations and then solve it:

   (i) $y' + xy = e^x y^{-n}, \ \ n \neq 0$ (ii) $y' + xy = x^2 y^5$ (iii) $xy' + y = x^4 y^3$ (iv) $xdy + ydx = xy^2 dx$

   (v) $xy^2 y' + y^3 = x \cos x$ (vi) $y(2xy + e^x)dx - e^x dy = 0$ (vii) $\sin y \dfrac{dy}{dx} = \cos x(2\cos y - \sin^2 x)$

   (viii) $\dfrac{dy}{dx} - \dfrac{\tan y}{1+x} = (1+x)e^x \sec y$

9. Solve the following by treating x as dependent variable and y as independent variable:

   (i) $(e^y - 2xy)y' = y^2$ (ii) $xy' + 2 = x^3(y-1)y'$ (iii) $y - xy' = y'y^2 e^y$ (iv) $y^2 + (x - 1/y)y' = 0$

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